

A Falsification-Ready Method for the $K3 \times T^2$ Reading of the Cooper s_7/s_{10} Families: Exact Computations, Formal Verification, and Controls

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Stream 2 — Selection & Geometry

(with the simulator stream's ledger, as reported by that stream)

2026-10-10 (preprint; not reviewed)

Scope

This report is about method and exact mathematics. Its object is the Shioda–Inose reading of $K3 \times T^2$ for two families of $K3$ surfaces. Results are stated at the tier of their evidence. The physical reading of $K3 \times T^2$ as a compactification, and any dark-sector observable, are Tier C — blocked physics and are outside what is computed here.

Abstract

We describe, and exercise on two families, a method for turning a geometric hypothesis into claims that can fail. Each claim is reduced to a statement an exact checker or a proof kernel can decide; each checker ships controls that must fail on tampered input; every number in a paper is read from a certificate by a script that refuses on drift; and a failing certificate has a stated consequence. Applied to the Shioda–Inose reading of $K3 \times T^2$ it gives the following. The transcendental lattice of $E \times E'$ for a cyclic n -isogeny is computed exactly and is isometric to $U \oplus \langle 2n \rangle$ at every level $n \leq 24$ and to no other $U \oplus \langle 2m \rangle$ (24 of 24 on the diagonal, 0 of 552 off it). It equals the certified lattice of each family at $n = 7$ and $n = 10$. The level read from the lattice agrees with the level read from the modular side. At $z = 1/27$ the section required by the M_7 polarization is constructed exactly, reconstructed from 52 specializations by an over-determined fit, and read: $\bar{P} \cdot \bar{O} = 5$, contact 0 with the A_1 fibre, height 14, equal to the lattice value. We list what would falsify each claim and how the pipeline is re-pointed at a new candidate by changing one integer.

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1 The approach

The programme’s hypothesis has a mathematical core (a family of $K3$ surfaces with a symmetric-square structure, a lattice, a modular coordinate) and a physical superstructure. The method keeps the two apart and makes the core falsifiable. It has six parts.

1. **Every claim carries its tier.** Kernel-checked statements (Tier A), exact derivations (Tier B), literature (Tier L) and blocked physics (Tier C) are never written in one another’s language. A prose lint applies the same forbidden-verb and conjecture rules to the whole body of each paper, and blocks fibre-type tokens that the programme’s ruling forbids.
2. **Numbers are read, not typed.** Each table is generated from a certificate by one script. Its `-check` mode exits non-zero if a fragment drifts, and its controls tamper with the certificates and require the table to change.
3. **Exact arithmetic, over-determined.** Lattices, intersection numbers and heights are computed in exact integer or rational arithmetic. Where a function must be reconstructed from specializations, the number of points exceeds the number of unknowns, the solution must be unique, and further points are held out and tested.
4. **Controls that must fail.** Every checker ships negative controls: a wrong level, a non-cyclic isogeny, a reversed orientation, a tampered lattice, a wrong normalization, too few points, a different section. A control that cannot fail is not a control.
5. **Formal verification where it reaches.** The symmetric-square factorization, the action of $\Gamma_0(N)^+$ on $U \oplus \langle 2N \rangle$ and the rank-jump lattice rows are kernel-checked in Lean 4 by the theory stream, with a second independent development for six of the rows. Release gates (build, absence of `sorry`, axiom audit, statement lock, producer $\neq$ verifier) are run by a session that did not write the code. The recorded rule is that the claim must live in the statement: a theorem can be true and compile while proving less than its name says.
6. **Removal and correction rules.** A failed hard gate removes a candidate; an absent certificate does not; consistency gates never score; and any correction is dated and machine-visible, not a footnote.

Result	Tier	Source of the number
$T(E \times E') \cong U \oplus \langle 2n \rangle$, $n = 7, 10$	B (exact computation)	<code>K3xT2_READING_S.json</code>
The same, for every $n \leq 24$ against every target	B	<code>K3xT2_READING_S_LEVEL_SWEEP.json</code>
$T(E \times E)$ at the three s_7 points of $\rho = 20$	B, forced	<code>K3xT2_READING_S.json</code>
Level consistency, lattice $n =$ modular n	B (hard gate)	<code>T3_LEVEL_CONSISTENCY.json</code>
The section at $z = 1/27$: $\bar{P} \cdot \bar{O}$, contact, height	B (exact; construction L)	<code>TW2_SECTION_DESCENT.json</code>
Counts of the simulator stream’s cross-check ledger	mirrored, not verified here	<code>refs/simulator_k3t2_ledger_counts_v3.json</code>
Identification of the T^2 of a compactification with E	C, not made	—

Table 1: What this report computes and where each number comes from. B: derived, not measured. L: literature. C: blocked physics.

2 Results

2.1 The lattice of $E \times E'$, and a sweep over levels

Let $E \times E' = \mathbb{C}^2/\mathbb{Z}^4$. Then $H^2(E \times E', \mathbb{Z}) \cong \wedge^2 \mathbb{Z}^4$ with the pairing given by the coefficient of $e_1 \wedge e_2 \wedge e_3 \wedge e_4$ in $u \wedge w$, and a two-dimensional subtorus spanned by v_1, v_2 has class $v_1 \wedge v_2$. For curves without complex multiplication (an assumption that is stated, not computed), the Néron–Severi lattice is spanned by the two fibres and the graph of the isogeny. The checker verifies signature $(1, 2)$ and saturation, computes the orthogonal complement as an integral basis, and finds an explicit unimodular basis change onto the target.

Family	n	NS sig.	saturated	$T(E \times E')$ (Gram), sig.	$\cong$ certified T
cooper_s7	7	(1, 2)	yes	$[0, 0, -1; 0, 14, 0; -1, 0, 0], (2, 1)$	yes
cooper_s10	10	(1, 2)	yes	$[0, 0, -1; 0, 20, 0; -1, 0, 0], (2, 1)$	yes
s_7 point	model j	CM disc.	$T(E \times E)$	certified T	agree
1/27	16581375	-28	$[2, 0; 0, 14]$	$[2, 0; 0, 14]$	yes (forced)
-1	-3375	-7	$[2, 1; 1, 4]$	$[2, 1; 1, 4]$	yes (forced)
∞	0	-3	$[2, 1; 1, 2]$	$[2, 1; 1, 2]$	yes (forced)

Table 2: Top: the computed $T(E \times E')$ against each family’s certified T . Bottom: $T(E \times E)$ at the s_7 points of Picard number 20, with the CM order identified by computation from the model’s j . The bottom agreement is forced (an elliptic point of the modular curve is a CM point), so it is a consistency check.

Level sweep (exact; assumes no CM)	Result
levels run	$1, \dots, 24$
(computed lattice, offered target) pairs	576
isometric on the diagonal (same level)	24 of 24
isometric off the diagonal (wrong level)	0 of 552
Néron–Severi of signature $(1, 2)$ and saturated at every level	yes

Table 3: The same code at every level: each computed lattice is compared with every target $U \oplus \langle 2m \rangle$. The isometry exists exactly when $m = n$.

The sweep is the evidence that the two carried levels, 7 and 10, are not special cases: a wrong level is rejected by the code that accepts the right one, at every one of the 576 pairs.

2.2 Level consistency

Family	n from lattice	n from modular side	Verdict
cooper_s7	7	7	T3_AGREE($n=7$)
cooper_s10	10	10	T3_AGREE($n=10$)

Table 4: The level n read from the lattice certificate against the level read from the modular side (an eta-quotient Hauptmodul with Newman’s conditions and an Atkin–Lehner check). Both legs start from the same operator, so agreement is a consistency check and not an independent measurement.

2.3 The section at $z = 1/27$

For $z = 1/27$ the section is the one attached to the degree-7 endomorphism $\sqrt{-7}$ of E . It is constructed, not inferred:

s_7 member	$ \text{disc } T $	ρ	extra roots	MW rank	$h(P)$	$\bar{P} \cdot \bar{O}$
generic	14	19	—	1	14	5
$z = 1/27$	28	20	A_1	1	14	5
$z = -1$	7	20	A_1	1	7/2	0
$z = \infty$	3	20	A_2	0	—	no section

Table 5: The generic member and the three $\rho = 20$ points of s_7 (lattice arithmetic and exact reduction of the model). At $z = \infty$ the Mordell–Weil rank is 0.

1. The Vélú isogeny $\varphi: E_1 \rightarrow E_2$ of degree 7 is written over $\mathbb{Q}$, and $\varphi_y = \varphi'_x$ is checked exactly against the curve equation.
2. The cubic C_t of Kumar–Kuwata (3.1) is reduced through $s = x_2 - t^2 x_1$ to a quartic with a rational point and mapped to a Weierstrass model by Mordell’s formulas, verified by identity on the actual divisor points and by the quartic’s own invariants.
3. The nine-point divisors $D^\pm$ ($\varphi_y(x_1) = \pm t^3$) are summed by Riemann–Roch: the function in $L(10O)$ vanishing on $D^\pm$ has one further zero R , and $Q^\pm = -R$. The section is $P = Q^+ - Q^-$.
4. The model is matched to the simple normal form $Y^2 = X^3 - 3\alpha X + t_s + t_s^{-1} - 2\beta$, with $t_s = t^6/343$ and $343 = \sqrt{\Delta_{E_1}/\Delta_{E_2}}$ computed; the j -match and the twist constant are asserted at every specialization.
5. $X'(t_s)$ is reconstructed as a rational function and read.

Quantity	Value
fit points / unknowns / held out	44 / 26 / 8
nullspace dimension (1 = unique)	1
$\deg N$, $\deg D^2$ of $X' = N/D^2$	14, 10
held-out points agree	yes
$\bar{P} \cdot \bar{O}$ (read)	5
contact with the A_1 fibre, contr (read)	0
height $h = 2\chi + 2\bar{P} \cdot \bar{O} - \text{contr}$ (read)	14
earlier steps and literature, read from their certificates	step 0: $h = 14$; step 2a: $\bar{P} \cdot \bar{O} = 5$, $\text{contr} = 0$; Kumar–Kuwata: $h = 14$
discriminant orders (model; fibration certificate)	$\{10, 10, 2, 1, 1\}$; $\{10, 10, 2, 1, 1\}$
cross-check agrees	yes

Table 6: The reconstruction and what is read from it. The cross-check row compares with the lattice steps and the fibration certificate, each read from its own file.

Result

The section meets O five times: once inside the A_1 fibre (so on its identity component) and four times at the roots of an irreducible quartic. The two E_8 fibres carry no contact. Its height is $14 = 2 \cdot 7$, equal to the value from the lattice arithmetic and from the Kumar–Kuwata theorem. Tier B — derived, not measured .

Remark 2.1. A control found a structural fact: $\bar{P} \cdot \bar{O} = (\deg N - 4)/2$ for any denominator, since poles that leave the finite line reappear at infinity. The count is fixed by the degree of the

numerator; the pole locations carry the contact information. Doubling the section does not fit the same shape, as expected from its height.

The section is a height-14 element; whether it generates the full Mordell–Weil group is the one question left open by this computation (no index is computed; the lattice and the literature both give 14).

2.4 The simulator stream’s ledger

The DualScaleSimulator stream compares, for 332 sealed targets of the Lean library, values from a separately written pipeline with the values stated in Lean. We mirror its counts, recounted from its row data by a script that refuses on mismatch and pinned to the source file’s hash. We report them as that stream’s record; the rows are not re-derived here.

Count in the simulator stream’s ledger	Value
sealed targets compared	332
AGREE	68
SAME_DECLARED_INPUT (downgraded)	3
NOT_COMPUTED	261
DISAGREE	0
rows with two or more independent routes	6
declared inputs (of which typed from memory)	80 (20)
rigidity entries: confirmed / disputed / unreviewed	3 / 2 / 28
chain links consistent; independent corroborations	8 (all); 3
items its authors could not do	30

Table 7: The simulator stream’s ledger, mirrored (source commit and hash pinned in `refs/`). “Rigid” there means a selecting condition that never mentions the target value was scanned for one parameter.

Most targets are not computed, so the ledger is a map of what remains. A fifth of the inputs it declares were typed from memory, which the ledger records. The same stream’s note records that its implemented model sits at level 12 through a hard-coded constant, that under the potential actually integrated no observable depends on the level, that an opt-in modular potential does read the level, and that its level experiment (item L5) ran only partially.

3 Falsification and re-adaptation

The value of the method is that a change of hypothesis is a change of input, and that each claim has a named way to fail.

Re-pointing the pipeline. A new candidate needs: its certified lattice (from the lattice checker); its level from the modular side; the level-consistency gate; and, if its lattice carries a Picard-number-20 point, the section construction at that point, which takes the isogeny degree and the curve as input. The sweep table reports that the lattice step runs unchanged at every level tried. A failed gate has a stated consequence (removal, or a dated correction), and every table in a new paper is regenerated by the same renderer, whose controls fail if a number is typed.

What the method can and cannot decide. It decides the mathematical core. It does not by itself connect that core to physics: that connection needs a worked effective-theory matching, a quantitative prediction, a pre-registered pin and a test against public data, and none of those links exists yet. The method is the same one that would be applied to them.

Claim	Would be falsified by	Exercised by
$T(E \times E') \cong U \oplus \langle 2n \rangle$ at level n	a level where the computed lattice is not isometric to its target, or is isometric to a different target	the 24-level sweep: 576 pairs, diagonal only; controls: wrong n , non-cyclic isogeny, reversed orientation, tampered Gram
A family's certified lattice has the level its modular coordinate has	a certificate with lattice $n \neq$ modular n (removes the candidate)	four other register entries fail the modular leg; 25 controls
The M_7 polarization needs a section of height $2n$ with $\bar{P} \cdot \bar{O} = n - 2$	no such section, or one of a different height, at a locus where the lattice allows it	the explicit construction at $z = 1/27$; the rank-0 locus $z = \infty$ has none, as the lattice step says
An isogeny of degree d gives a section of height $2d$	a graph whose reconstructed height differs	the doubled section fails the shape; 14 controls
A lattice or rank-jump statement of the theory	a failing gate of the Lean release run, or a statement that omits what its name claims	kernel build, axiom audit, statement lock, producer $\neq$ verifier

Table 8: What would falsify each claim, and what already exercises it.

4 Corrections recorded in band

Corrections recorded in band

1. The reading of “ $K3 \times T^2$ ” used here was first described as the programme’s choice before it had been ruled. It was ruled on 2026-10-08 (decision D29’, taken on the decision authority’s behalf under explicit delegation) and is a convention for selection criteria only.
2. The first reconstruction of the section used a transcription of the Weierstrass form from a garbled line of the source. The exact j -match test refused it, and the source’s simple normal form was used; the constant $t_s = t^6/343$ was calibrated on three points and is asserted at all fifty-two.
3. A first sensitivity control for the read-off passed for the wrong reason (removing the quartic moves its poles to infinity, leaving the total unchanged). It was replaced by one that adds a genuine pole.
4. A note elsewhere in the programme said the host lacked a numerical library and that two test modules were ignored. That was a property of the system interpreter; under the project’s own environment the full suite runs.
5. One continuous-integration run of the simulator failed once on a 10^{-12} tolerance in an ODE-solve comparison and passed on re-run of the same commit; the tolerance is hardware-sensitive and was not changed.

5 Reproducibility

Every number in tables 2 to 7 is read from a certificate or a pinned mirror by `scripts/render_paper_tables.py`. Checkers and their controls:

- `checkers/check_K3xT2_reading_S.py` (9 controls);
- `checkers/check_K3xT2_reading_S_level_sweep.py` (6 checks);

- `checkers/check_T3_level_consistency.py` (25 controls);
- `checkers/check_TW2_section_descent.py` (14 controls);
- `scripts/extract_simulator_ledger_counts.py` (recount; 8 controls).

Decision records: `briefs/T0_DECISIONS_2026_10_07_STREAM2_DELEGATED.md` (D29'–D31');
Lean release-gate record: Stream 1 `briefs/STREAM2_REVERIFICATION_2026_10_08.md`.

Provenance

Generated-by: Claude (Fable 5.1 and Sonnet 5.5), Stream 2 | Verified-by: `render_paper_tables.py`–`check` and its controls; each cited checker's own controls | Reviewed-by: N (preprint; not reviewed).

References

- [1] A. Kumar and M. Kuwata, “Elliptic $K3$ surfaces associated with the product of two elliptic curves,” *arXiv:1409.2931*.
- [2] M. Schütt and T. Shioda, “Elliptic surfaces,” *arXiv:0907.0298*.
- [3] T. Takatsu, “On the geometry of singular $K3$ surfaces with discriminant 3, 4 and 7,” *arXiv:1903.03054*.
- [4] I. Dolgachev, “Mirror symmetry for lattice polarized $K3$ surfaces,” *arXiv:alg-geom/9502005*.
- [5] X. Callens and the SocrateAI Scientific Agora Collaboration, “The selected $K3$ surfaces of the Cooper s_7/s_{10} families: identification, the $K3 \times T^2$ reading, and the twisted-Weierstrass route,” Stream 2 note, 2026-10-08.